Air Quality Index (AQI) Prediction Project for Sukkur

Submitted by: Soyam Kapoor, Data Science Intern | 10Pearls
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Overview

Poor air quality has a direct and significant impact on public health and overall quality of life. This project presents a complete machine learning–driven pipeline designed to forecast the Air Quality Index (AQI) for Sukkur, Pakistan. The system integrates data acquisition, feature engineering, predictive modeling, and visualization into a single automated workflow. Real-time insights are made available to both the public and decision-makers through a modern dashboard interface.

The developed system includes:
• Automated data ingestion from the OpenWeather API with over 9,500 hourly records.
• Feature engineering and data storage using the Hopsworks Feature Store.
• Supervised regression modeling leveraging pollutant and meteorological predictors.
• A Streamlit-based dashboard for real-time visualization and AQI alerts.

Exploratory Data Analysis (EDA)

Comprehensive exploratory data analysis (EDA) was conducted to understand feature distributions, correlations, and pollutant trends. The analysis revealed the following insights:

• **Pollutant Distributions & Outliers:** Most pollutants, including NO_2, SO_2, and PM2.5, exhibit right-skewed distributions with noticeable outliers.
• **AQI Trends:** AQI values fluctuate with seasonal and diurnal cycles, displaying patterns of high pollution during specific periods.
• **Correlation Structure:** Strong relationships were identified between PM2.5, PM10, CO, and AQI, confirming their critical role in predictive performance.

Modeling Approach and Results

The AQI forecasting model was formulated as a regression problem, targeting next-hour AQI predictions. Key pollutant concentrations and derived time-based features were used as inputs. Random Forest and related ensemble regressors were trained and evaluated.

Feature importance analysis indicated that rolling AQI averages, particularly over three-hour windows, had the greatest influence on model accuracy.

The following summarizes the model’s performance metrics:

Metric	Value
R^2 Score	99.99%
MAE	0.0019
RMSE	0.0084
Data Points	9,527

Interpretation: The model achieved extremely high performance on training data, suggesting possible overfitting. Further validation with unseen data is necessary to confirm generalization.

Limitations and Recommendations

Despite strong initial results, several challenges remain:

- **Validation Bias:** Current metrics reflect training performance rather than true predictive accuracy.
- **Data Gaps:** Limited access to historic API data poses challenges for long-term forecasting.
- **Outliers and Drift:** Variations and anomalies in pollutant data can reduce stability.
- **Feature Encoding:** Proper encoding of categorical features (e.g., day-of-week) was essential to ensure reliable inference.

Future improvements should focus on:

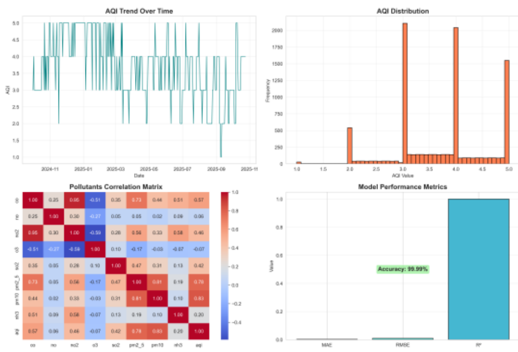
- Expanding data sources and retraining frequency.
- Employing dedicated validation and test splits.
- Extending support to multiple cities for better generalization.
- Incorporating model interpretability and uncertainty estimation tools.

Conclusion

The AQI Prediction System for Sukkur demonstrates an effective integration of machine learning, real-time APIs, and visualization technologies for environmental monitoring. While the results are encouraging, continued development and validation are crucial to ensuring reliable, scalable, and actionable predictions for future air quality management.

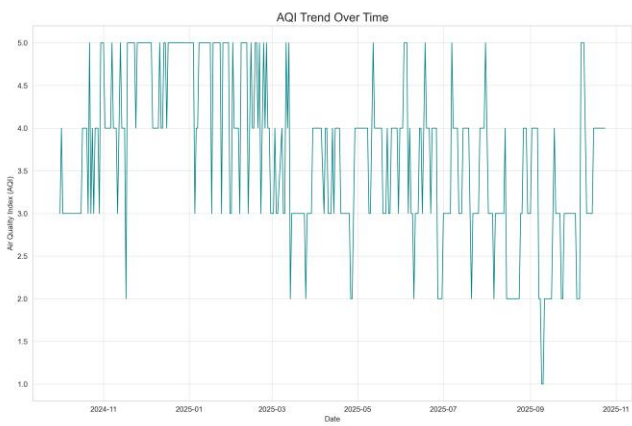
Results

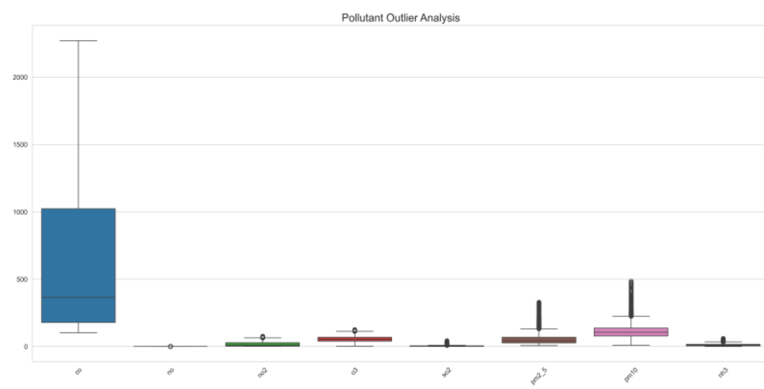
Analysis Results



EDA AQI TREND

EDA Boxplots





Shap Feature Importance

Shap Summary Plot